

Glossary & Machine Context

LG Electronics Noida — AI Training. Plain-language reference for the terms used today.

The machine at the centre of everything: the IM-500.

The **IM-500** is a plastic **injection-molding machine** on the Bay-3 production line. It melts plastic pellets and injects the melt into a mould under high pressure to produce plastic parts. Like any heavy machine it wears out, and a surprise breakdown stops the whole line. It carries **sensors** — vibration, oil and barrel temperature, motor current, screw torque, tie-bar strain, and servo position — that reveal its health. Our whole example is about reading those sensors to **predict** a failure, **find the spare part**, and **schedule maintenance before it breaks**. Everything we build today serves that one goal.

Machine & maintenance terms

Bay-3

The production bay (area of the factory floor) where the IM-500 sits, with its own spare-parts store.

Sensor reading

A live number from the machine (e.g. vibration in mm/s). Rising trends warn of wear.

Predicted failure / lead time

How long until a part is expected to fail (e.g. "~6 days"). Maintenance must be scheduled before this date.

Spare part / part number

The replacement component and its code (e.g. **EN-77** = servo encoder, **BR-3105** = pump bearing).

In stock / min stock

How many spares are on the shelf, and the minimum we should keep. Below minimum = reorder.

Supply lead time

How long the supplier takes to deliver a part (e.g. EN-77 = 21 days). If this is longer than the failure lead time, it's urgent.

EN-77

The star of the demo: a servo encoder that's out of stock (0), with a 21-day supplier lead time but only ~6 days to failure — so it must be escalated urgently.

URGENT / NORMAL

Priority of a maintenance request. URGENT = part missing or won't arrive in time; escalate to the Bay-3 lead.

AI concepts

AI (Artificial Intelligence)

Software that follows rules or spots patterns — e.g. "flag if vibration > 4.5". No language, no reasoning.

Generative AI (GenAI)

AI that creates language or content — e.g. writes a shift report from notes. On its own it doesn't know your machine.

LLM (Large Language Model)

The "brain" behind GenAI — a model trained on huge text that understands and writes language (e.g. Google Gemini).

AI Agent

An LLM given reference data and the ability to decide/act — reads an alarm, looks it up, decides an action. (Our Agent A/B/C.)

Agentic AI

Several agents coordinating on a task — predict → check parts → raise maintenance. A team, not a single bot.

Orchestration

Wiring agents together and deciding who hands off to whom. We use a **sequential** pipeline: A → B → C.

Handoff

One agent passing its result as the input to the next agent.

RAG (Retrieval-Augmented Generation)

Give the AI your documents so it answers **ONLY** from them, with citations — instead of guessing. Stops "hallucination".

Hallucination

When an AI makes up a confident but wrong answer. RAG and grounding prevent it.

Grounding / citation

Tying an answer to a specific source document so it can be trusted and checked.

System prompt

The standing instructions we bake into an agent — its role plus the reference data it must use.

Structured output

Forcing the AI to answer in fixed fields (part, priority, days...) so the result can go straight into a spreadsheet.

Tools we use (and why)

Tool	What it is	Why we use it
NotebookLM	Google's no-code RAG tool	Upload PDFs, ask grounded questions — free, stable, zero setup.
n8n	No-code visual workflow builder	Build and connect the agents by dragging boxes — free tier, no licence.
Google Gemini	The AI model (LLM)	Free API tier, strong, instant key from Google AI Studio.
	Spreadsheet	

Google Sheets	Where the agent logs each maintenance decision — a permanent, shareable record.
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n8n building blocks

Workflow

One automation on one canvas — e.g. our $A \rightarrow B \rightarrow C$ pipeline.

Node

One step/box on the canvas, wired left to right.

Trigger

The first node that starts the workflow — we use a chat box ("Chat Trigger").

AI Agent node

The box that holds an agent; you plug a Chat Model (the LLM) into it.

Credential

A saved login/API key, stored once and reused (e.g. the Gemini key, the Google login).

API key

A secret code that lets one tool (n8n) use another (Gemini).

Rate limit / quota

How many free requests you get per day. Use a stable model (~1,500/day), not a preview (~20/day).